10.1 Hypothesis Testing - Null and Alternative Hypothesis.

Source: The Organic Chemistry Tutor - https://www.youtube.com/watch?v=wiYJWyfdGg4&list=PL0o_zxa4K1BVszIlRdfv4Hl4UIqDZhXWV&index=71

1. Company XYZ manufactures calculators with an average mass of 450 g. An engineer believe that average weight to be different and decides to calculate the average mass of 50 calculators. State the null and alternatives hypothesis?

Null Hypothesis : H_0

↳ Status quo, claim, current accepted value that the majority of people believe its true.

Alternative Hypothesis : H_a

↳ Contrast, contradictory to null hypothesis, this is what researcher try to prove in order to disprove the null hypothesis. → Fail to prove H_a → Null hypothesis likely to be correct.

Null : H_0 $\quad \mu = 450\ g$

Alternative : H_a $\quad \mu \neq 450\ g$

2. The teacher in a school believe that at lessa 80% of student will complete high school. A student disagree with this value and decide to conduct a test. State the null and alternatives hypothesis.

Null : $H_0 : P(A) \geq 80\%$ $\quad$ A = student complete high school.

Alternatives : $H_a : P(A) < 80\%$

1. **As an event:** let A = "a randomly selected student completes high school".
   - H_0: P(A) ≥ 0.80
   - H_a: P(A) < 0.80
2. **As a 0/1 random variable:** let X = 1 if a randomly selected student completes high school, and X = 0 if not. X follows a Bernoulli distribution with probability p.
   - H_0: P(X = 1) ≥ 0.80
   - H_a: P(X = 1) < 0.80

3. A teacher wishes to test if the average of GPA of students in the highschool is different from 2.7. State the null and alternatives hypothesis.

$H_0 : \mu = 2.7$

$H_a : \mu \neq 2.7$

10.2 How to Identify Type 1 and Type 2 Errors

Type I Error: Rejecting the null hypothesis when it is true.
Type II Error: Not Rejecting the the null hypothesis when it false.

$P(\text{type I error} \mid H_0 \text{ is true}) = \alpha$
$P(\text{type II error} \mid H_0 \text{ is false}) = \beta$
$P(\text{rejecting a false } H_0) = 1 - \beta$ → is also referred to as the power of the test

Decision	H_0 True	H_0 False
Reject H_0	Type I Error (False Positive)	✓ (True positive)
Fail to reject H_0	✓ (True Negative)	Type II Error (False Negative)

1. Let's say that null hypothesis H_0 is: John's used car is save to drive.

(a) John thinks that his car may be safe when, in fact, it is not safe.
↳ fail to reject ↳ false Type II FN
(b) John thinks that his car may be safe when, in fact, it is safe.
↳ fail to reject ↳ true VTN
(c) John thinks that his car may not be safe when, in fact, it is not safe.
↳ reject H_0 ↳ true VTP
(d) John thinks that his car may not be safe when, in fact, it is safe.
↳ reject H_0 ↳ false Type I FP

(a). Which statement represent Type I error (a)
(b). Which statement represent Type II error (d)
(c). Which type of error has greater consequences (Type II / FN / (a))